Trøndelagkonkurranse 2026 - Level I

Second Round - October 24, 2026

1st Part

Choose and solve only one of the following problems:

Problem 1

Find the second decimal place after the comma in the decimal expansion of

$$\frac{1}{1+\sqrt{101}+\sqrt{103}}$$

Note: The second decimal place after the comma in π is 4.

Problem 2

Frida writes each of the numbers $1, 2, 3, \ldots, 2026$ on a blackboard once. At each step, she can erase two numbers on the blackboard, say x and y, and replace them with the single number $|x-y|$. After several steps, there is only one number on the blackboard. Is it possible that she obtains 0 at the end?